

Data Analysis Report

Dataset: Project_1_-_Weather_Dataset.csv

<head>

<title>Comprehensive Data Analysis: Weather Dataset</title>

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<class 'pandas.core.frame.DataFrame'>

DatetimeIndex: 8784 entries, 2012-01-01 00:00:00 to 2012-12-31 23:00:00

Data columns (total: 7 columns):

This dataset captures various meteorological parameters, likely on an hourly basis, for a specific location over approximately one year. The dataset comprises 8784 records (rows) and 7 distinct features (columns). The features include:

1 Hum_% 8784 non-null int64

2 Temp_C 8784 non-null int64 Ambient Temperature in Celsius.

3 Wind_Speed_kmh 8784 non-null float64

4 Visibility_km 8784 non-null float64

5 Dew_Point_Temp_C 8784 non-null float64 Dew Point Temperature in Celsius.

6 Rel_Hum_% 8784 non-null float64 Relative Humidity as a percentage.

7 Weather 8784 non-null object

dtypes: float64(4), int64(2), object(1)

memory usage: 549.0+ KB

Visibility_km: Visibility in kilometers.

Press_kPa: Atmospheric Pressure in kilopascals.

Weather: A categorical description of the prevailing weather conditions.

Index	Temp_C	Dew Point Temp_C	Rel Hum_%	Wind Speed_kmh	Visibility_km	Press_kPa	Weather
2012-01-01 00:00:00	-1.8	-3.7	87	4	8.0	101.24	Fog
2012-01-01 01:00:00	-1.8	-3.4	89	7	4.0	101.26	Freezing Drizzle,Fog
2012-01-01 02:00:00	-1.5	-3.2	88	6	4.0	101.27	Freezing Drizzle,Fog
2012-01-01 03:00:00	-1.5	-3.0	88	7	4.0	101.28	Fog

<h3>? Data Quality Assessment</h3>

<p>Based on the provided summary statistics, the data quality appears to be generally Good.</p>

- Missing Values: All columns show a <code>count</code> of 8784, matching the total number of records. This indicates a complete dataset with no missing values across any of the observed features, which is excellent.

Constant/Near-Constant Columns: No columns appear to be constant or near-constant. All numerical columns exhibit a range of values (min/max) and non-zero standard deviations. The 'Weather' column has 50 unique values, indicating sufficient variability.

Data Types: The numerical columns (Temp_C, Dew Point Temp_C, Rel Hum_%, Wind Speed_kmh, Visibility_km, Press_kPa) are appropriately represented as floating-point numbers. The 'Weather' column is categorical/string, which is suitable for its descriptive nature.

Suspected Quality Issues: No immediate systemic quality issues (e.g., incorrect data types, obvious data entry errors) are apparent from the summary statistics.

Duplicate Records: Cannot be directly assessed from the

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provided summary statistics. However, given the complete counts for all columns, it's unlikely there are exact duplicate rows if the data is time-indexed and unique per observation point.

Index	Temp_C	Dew Point Temp_C	Rel Hum_%	Wind Speed_km/h	Visibility_km	Press_kPa
count	8784.0	8784.0	8784.0	8784.0	8784.0	8784.0
mean	8.79814435336976	2.55529371584699	67.4316939890710	14.9454690346083	27.6644467213114	101.051623406193
std	11.6827096912	10.8830733109557	18.908814700758	18.68269607946885	11.6225882451114	8.8410047459486
min	-28.5	-28.5	18.0	0.0	0.2	97.52
25%	0.1	5.9	36.0	9.0	24.1	100.56
50%	9.3	3.3	68.0	13.0	25.0	101.07
75%	18.8	11.8	81.0	20.0	25.0	101.59
max	33.0	24.4	100.0	83.0	48.3	103.65

Other numerical columns (Temperature, Humidity, Visibility) show ranges that are plausible for a year's weather observations, with maximums and minimums generally within 2-3 standard deviations from their respective means, suggesting less severe outlier behavior.

Overall Quality Rating: Good. The absence of missing data is a significant strength. While some extreme values exist in 'Wind Speed_km/h' and 'Press_kPa', these might represent valid, albeit rare, weather events rather than data errors. Further investigation into these extremes is recommended.

? Key Statistical Insights

- Temperature (Temp_C):**
 - The average temperature is **8.80°C**, with a median of **9.30°C**. This slight difference suggests a minor left-skew, indicating a slightly longer tail towards colder temperatures.
 - Temperatures range widely from a freezing **-23.30°C** to a warm **33.00°C**, reflecting a full annual cycle.
 - The standard deviation of **11.69°C** highlights significant variability throughout the year.
- Dew Point Temperature (Dew Point Temp_C):**
 - The mean dew point is **2.56°C**, with a median of **3.30°C**. Similar to ambient temperature, it shows a slight left-skew.
 - The range is from **-28.50°C** to **24.40°C**, generally tracking the ambient temperature but

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2. AI Analysis Summary

consistently lower.

Relative Humidity (Rel_Hum_%):

- The average relative humidity is **67.43%**, with a median of **68.00%**. This indicates a fairly balanced distribution.
- Humidity levels span from a very dry **18.00%** to a saturated **100.00%**.

Wind Speed (Wind_Speed_kmh):

- The mean wind speed is **14.95 km/h**, but the median is lower at **13.00 km/h**. This, coupled with a maximum of **83.00 km/h**, indicates a right-skewed distribution, meaning most observations have lower wind speeds, but there are occasional occurrences of significantly high wind speeds.
- The minimum wind speed is **0.00 km/h**, representing calm conditions.

Visibility (Visibility_km):

- Average visibility is **27.66 km**, with a median of **25.00 km**. The distribution is slightly right-skewed.
- Visibility ranges from a very low **0.20 km** (likely fog or heavy precipitation) to a maximum of **48.30 km**.

Atmospheric Pressure (Press_kPa):

- The mean pressure is **101.05 kPa**, with a median of **101.07 kPa**, suggesting a relatively symmetrical distribution around the average.
- Pressure varies between **97.52 kPa** and **103.65 kPa**, indicating the presence of both low-pressure (stormy) and high-pressure (stable) systems.

Weather Conditions (Weather):

- There are **50 unique weather conditions** recorded.
- The most frequent weather condition is **'Mainly Clear'**, occurring **2106 times**, which accounts for approximately **24.0%** of all observations (2106 / 8784). This suggests 'Mainly

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Clear' is a dominant weather pattern for this location.

<p>Notable Ratios & Patterns: The average difference between ambient temperature and dew point temperature is approximately 6.24°C (8.80°C - 2.56°C). This 'dew point depression' is a key indicator of atmospheric moisture content; a larger difference generally means drier air, while a smaller difference (approaching zero) indicates higher humidity and potential for fog or precipitation.</p>

<p>Surprising Finding: The maximum wind speed of 83.0 km/h stands out significantly from the typical wind conditions, being more than five times the average speed. This points to the occurrence of at least one, or a few, exceptionally strong wind events during the observed period.</p>

<h3>?? Temporal Patterns & Trends</h3>

<p>Not applicable (no datetime index).</p>

<h3>? Correlations & Relationships</h3>

<p>Direct correlation coefficients between variables cannot be calculated solely from the provided summary statistics (mean, std, min, max, quartiles). However, based on the nature of meteorological data, we can infer plausible relationships:</p>

Temperature and Dew Point Temperature: These are typically strongly positively correlated. As ambient temperature rises, the air's capacity to hold moisture increases, often leading to higher dew point temperatures, and vice-versa.

Relative Humidity, Temperature, and Dew Point Temperature: Relative humidity is inversely related to the difference between ambient and dew point temperatures. A strong negative correlation would be expected between relative humidity and the temperature-dew point spread. When ambient temperature and dew point are close, relative humidity is high.

Visibility and Weather: Certain 'Weather' conditions (e.g., 'Fog', 'Haze', 'Snow', 'Rain') would likely show a strong negative correlation with 'Visibility_km', meaning visibility decreases significantly during these events.

Wind Speed and Pressure: Often, periods of very low atmospheric pressure are associated with stronger winds and stormy weather, suggesting a potential negative correlation, though this relationship can be complex.

<p>Without specific correlation coefficients, it is difficult to distinguish between potential causal links and coincidences. For instance, high humidity (small Temp-Dew Point difference) is a causal factor for fog (low visibility), but high winds might coincide with low pressure without a direct causal link in all scenarios.</p>

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?? Anomalies & Irregularities

Extreme Wind Speed (83.0 km/h): This value is significantly higher than the 75th percentile (20.0 km/h) and the mean (14.95 km/h). It represents a severe wind event, potentially a gale or storm. Its severity is high, as it falls far outside the typical range of observed wind speeds. The potential cause is a powerful weather system passing through the area.

Extreme Atmospheric Pressure (Min: 97.52 kPa, Max: 103.65 kPa): These values represent periods of unusually low and high atmospheric pressure. Low pressure (97.52 kPa) is often associated with intense storms or depressions, while very high pressure (103.65 kPa) indicates strong, stable anticyclonic conditions. These are significant deviations from the mean (101.05 kPa) and suggest the occurrence of major weather fronts or systems.

No other clear anomalies (spikes, dips, gaps) can be definitively identified from the aggregated summary statistics alone. The minimum visibility of 0.2 km, while very low, is a plausible value for dense fog or whiteout conditions.

? Prognosis & Future Outlook

<p>Based on the observed patterns and distributions over what appears to be a full year of data, the following future directions are plausible:</p>

Expected Continuations:

The prevalence of 'Mainly Clear' conditions (24% of the time) suggests this will continue to be a dominant weather type. Confidence Level: High.

The annual cycle of temperatures, dew points, and humidity will likely repeat, with similar ranges and variability observed year-on-year. Confidence Level: High.

The occurrence of calm wind conditions (0 km/h) and moderate wind speeds (median 13 km/h) will remain frequent. Confidence Level: High.

Potential Risks:

The data indicates a recurring risk of severe weather events characterized by very high wind speeds (up to 83 km/h) and extreme low/high atmospheric pressure. These events, while infrequent, have the potential for significant impact. Confidence Level: Medium.

Periods of very low visibility (down to 0.2 km) due to fog or heavy precipitation are also a recurring risk, impacting transportation and outdoor activities. Confidence Level: Medium.

Growth/Decline Scenarios: Without a temporal index

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spanning multiple years, it is impossible to predict long-term growth or decline in any weather parameter (e.g., climate change trends). The dataset represents a snapshot of one year's typical variability. *Confidence Level: Low (due to data limitations).*

? Actionable Observations

Investigate Extreme Weather Events: Prioritize detailed analysis of the periods corresponding to the maximum wind speed (83.0 km/h) and the extreme pressure readings (97.52 kPa, 103.65 kPa). Understanding the specific 'Weather' conditions and other parameters during these times can provide critical insights into severe weather patterns and their potential impacts.

Enhance Data with Temporal Index: The most crucial improvement is to integrate a precise datetime index for each observation. This will unlock the ability to analyze seasonal cycles, daily patterns, long-term trends, and enable time-series forecasting, significantly enhancing the value of the dataset.

Categorical Weather Analysis: Conduct a deeper dive into the 'Weather' column. Analyze the frequency and duration of specific weather types beyond 'Mainly Clear', especially those associated with adverse conditions (e.g., 'Fog', 'Rain', 'Snow', 'Thunderstorms'). This can inform risk assessments and operational planning.

Explore Inter-Variable Relationships: Once a full dataset (not just summary statistics) is available, compute and visualize the correlation matrix between all numerical variables. This will empirically confirm hypothesized relationships (e.g., Temp vs. Dew Point, Humidity vs. Visibility) and potentially uncover unexpected dependencies.

?? Limitations & Caveats

Lack of Temporal Index: The most significant limitation is the absence of a specific datetime column. This prevents any analysis of trends over time, seasonality (e.g., monthly temperature averages), daily cycles, or time-series forecasting. The assumption of a full year of hourly data is based solely on the row count.

Summary Statistics Only: The analysis is based purely on aggregated summary statistics (`describe()` output). This limits the ability to perform detailed exploratory data analysis, such as visualizing distributions, identifying specific data points, or calculating direct correlations between variables.

Single Location Assumption: The data is assumed to be from a single meteorological station. Conclusions drawn may not be generalizable to other geographical locations.

Outlier Context: While extreme values were flagged, without the full dataset and temporal context, it's impossible to definitively determine if they are true data anomalies or valid records of rare, severe weather events.

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- "Weather" Column Granularity:** While 50 unique weather types are present, the specific definitions or groupings of these conditions are unknown, which could impact the interpretation of 'Weather' patterns.

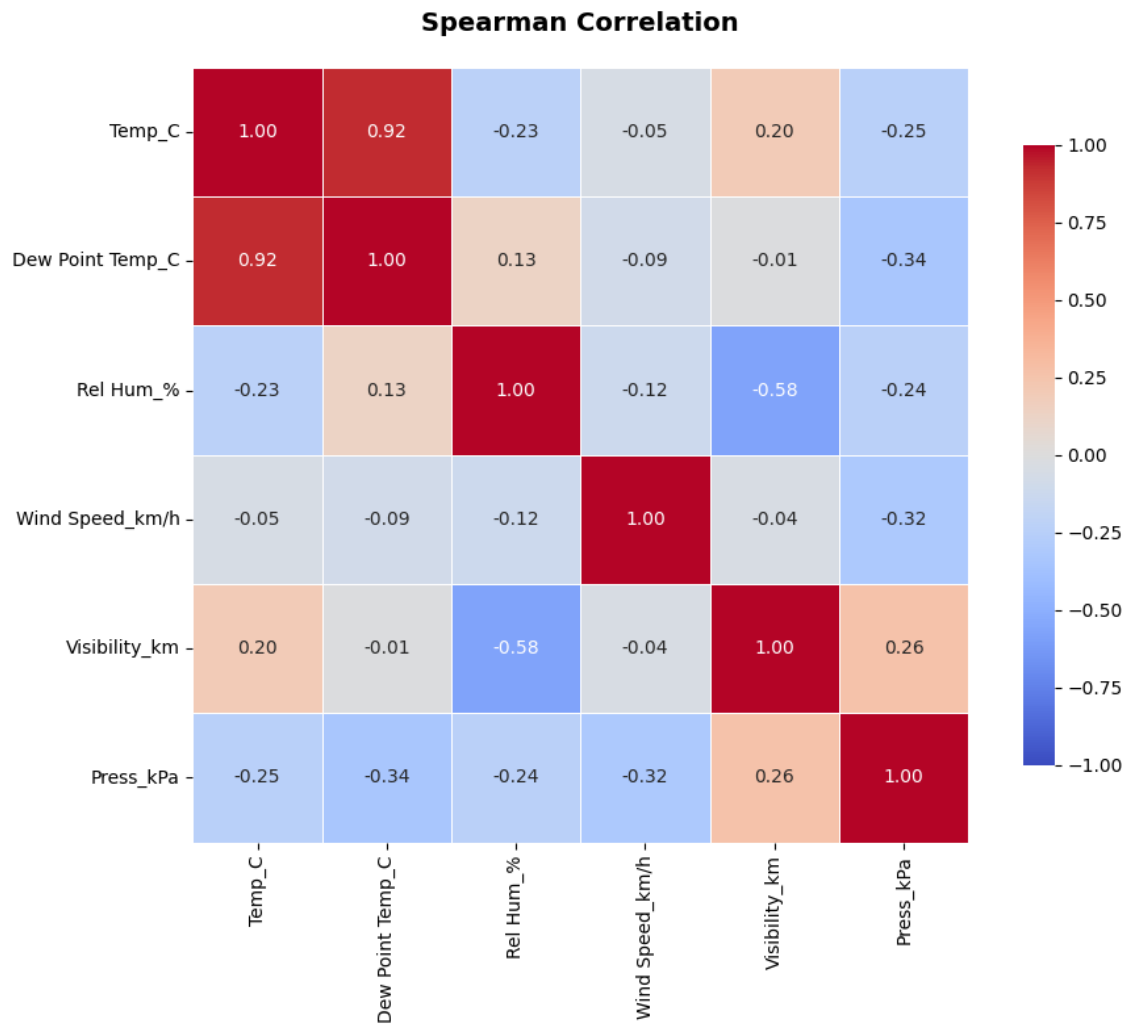
- No External Data:** The analysis is confined to the provided dataset. No external factors (e.g., geographical features, urban heat island effect, climate change data) were considered, which could influence weather patterns.

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3. Correlation Analysis

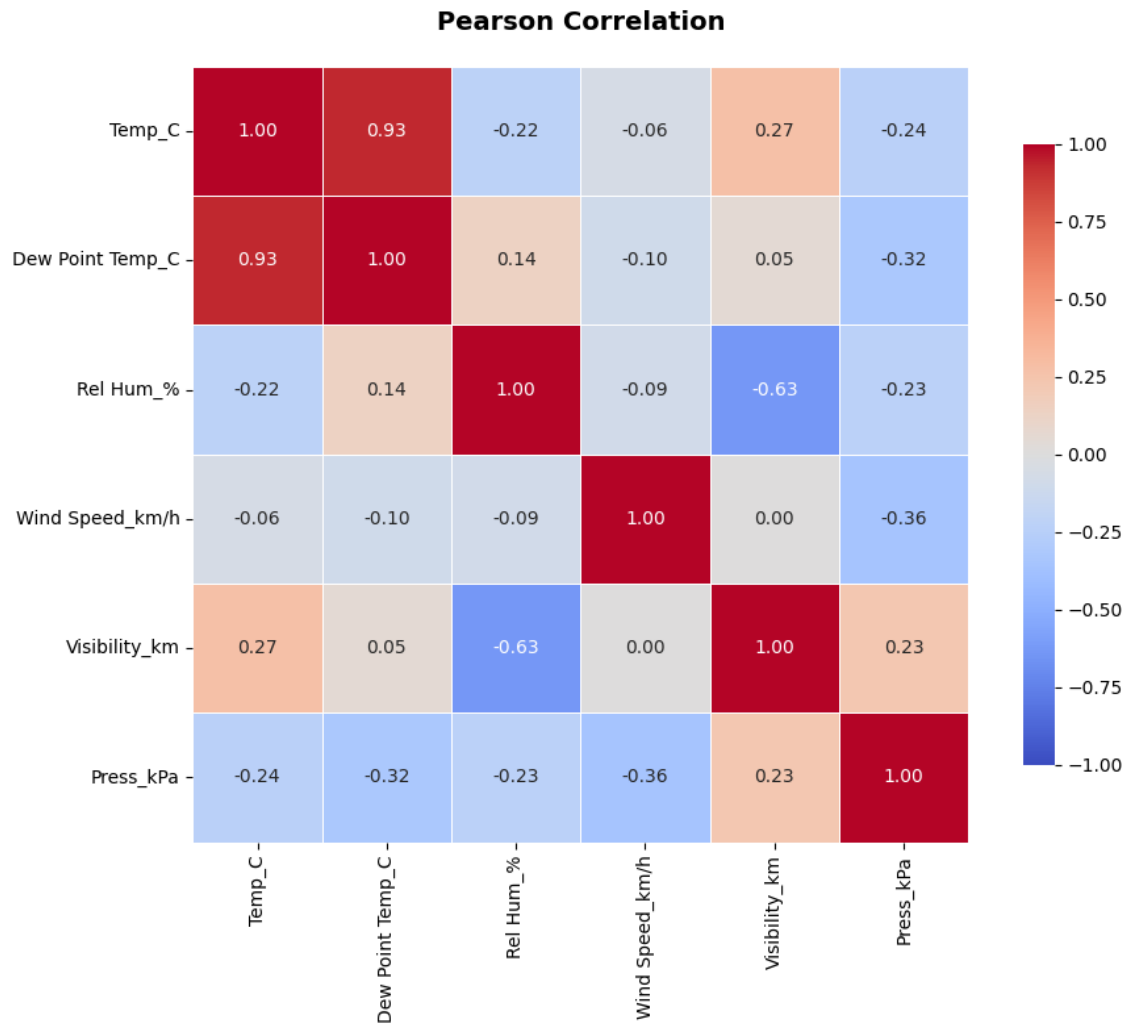
Spearman Correlation:



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Pearson Correlation:



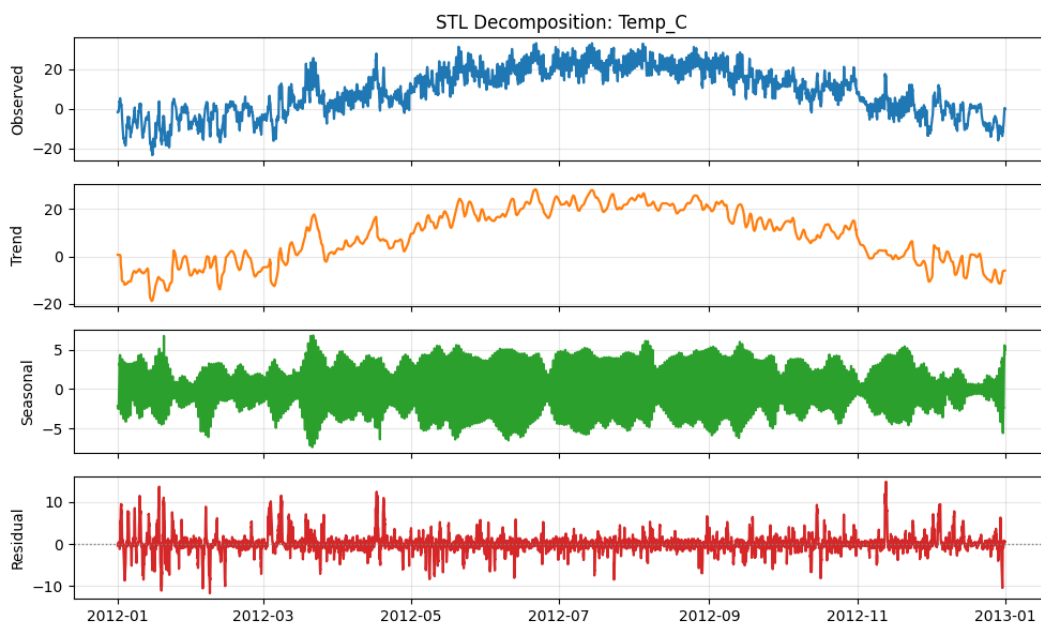
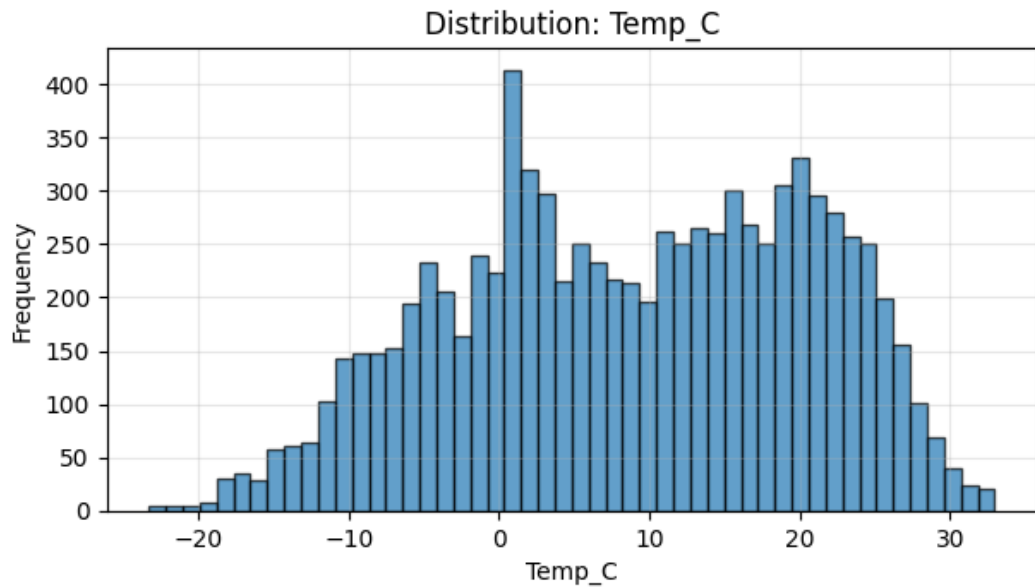
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4. Column Analysis

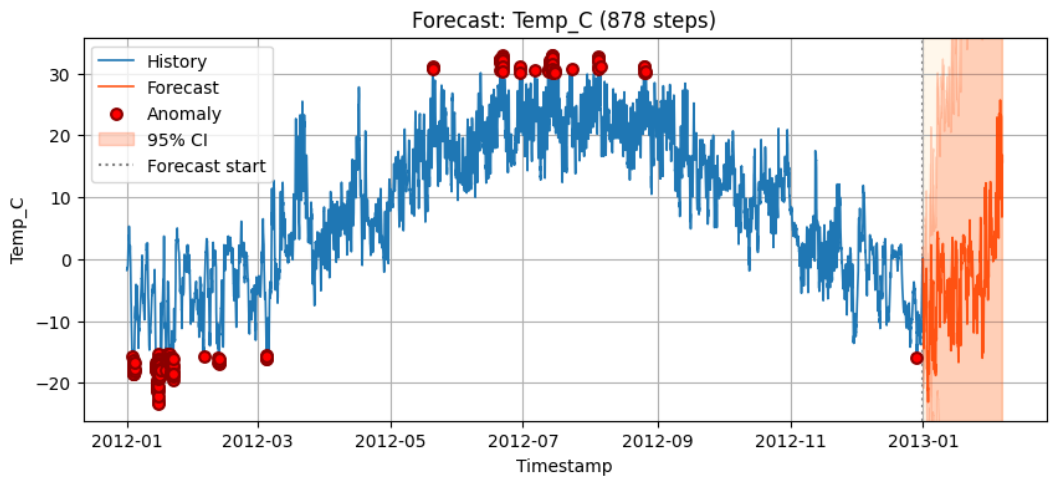
Column: Temp_C

Column: Temp_C



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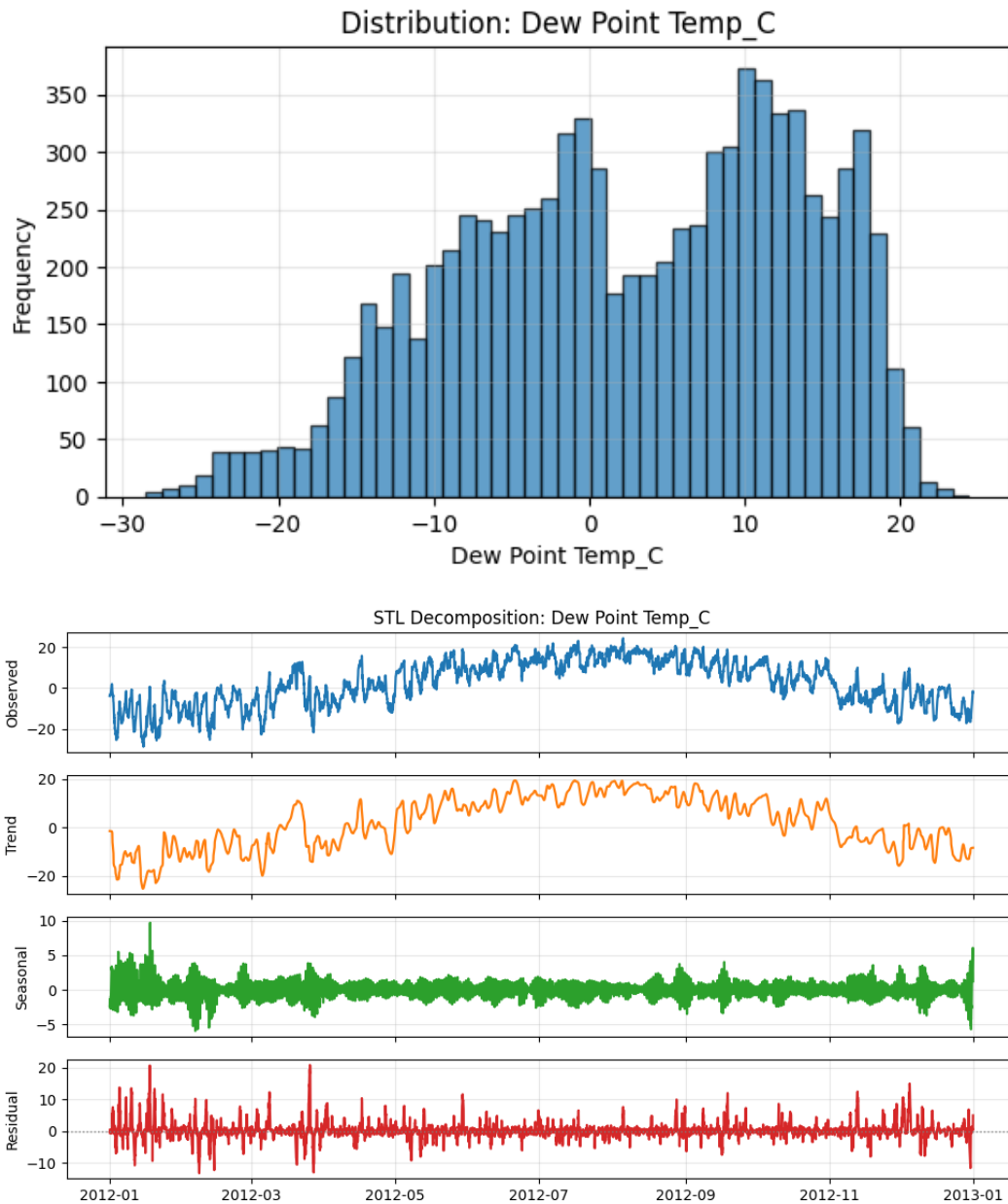


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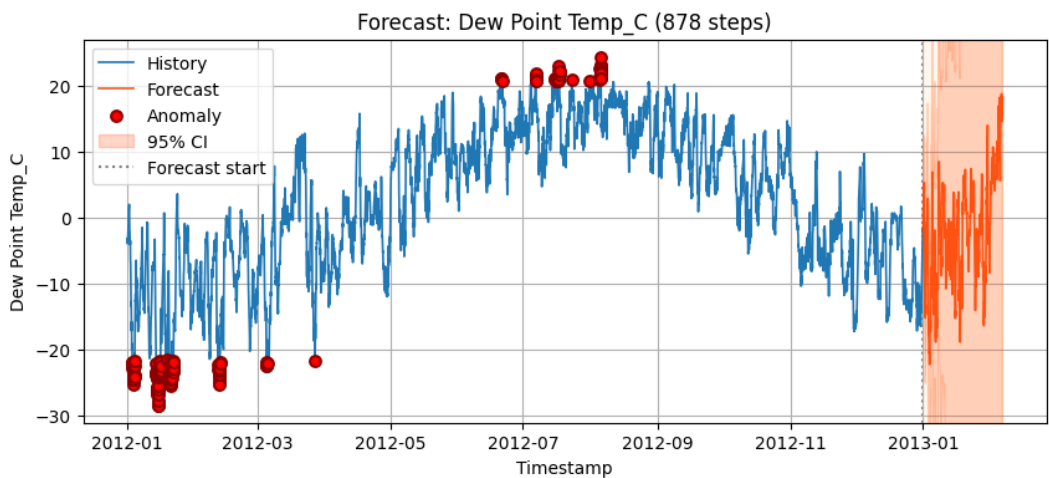
Column: Dew Point Temp_C

Column: Dew Point Temp_C



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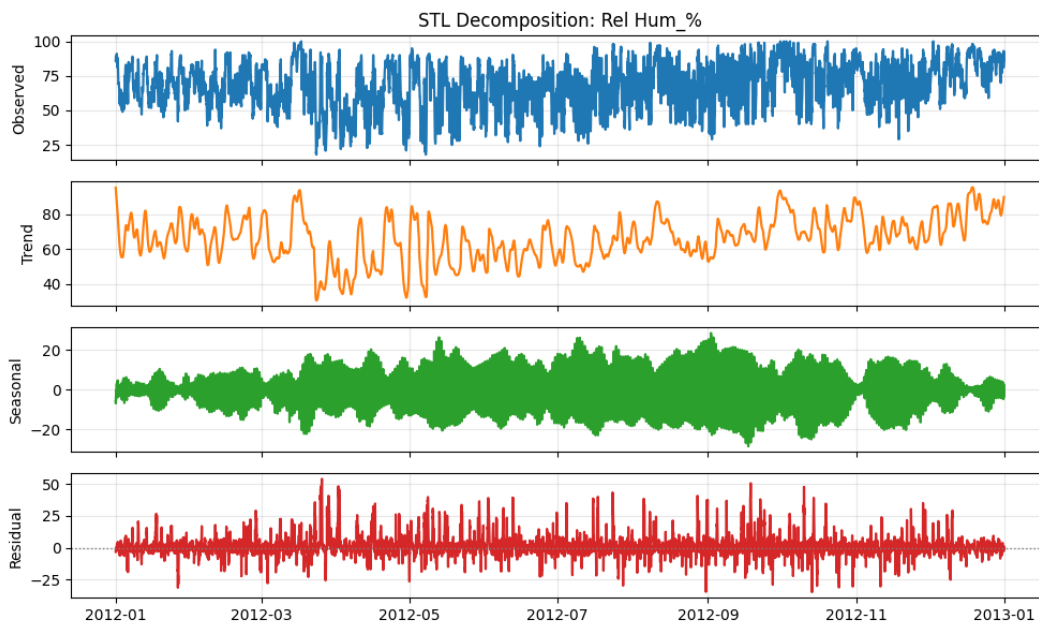
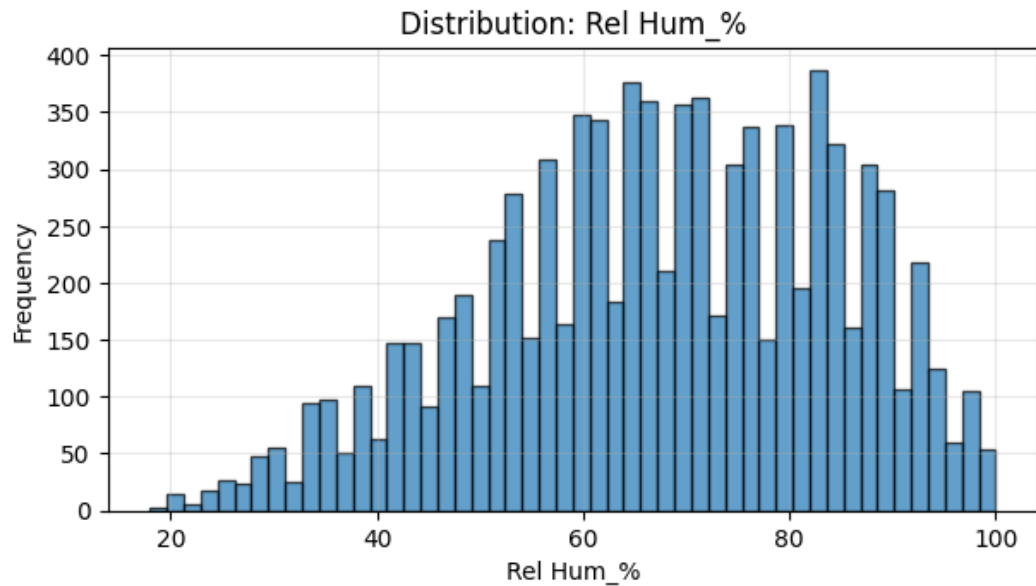


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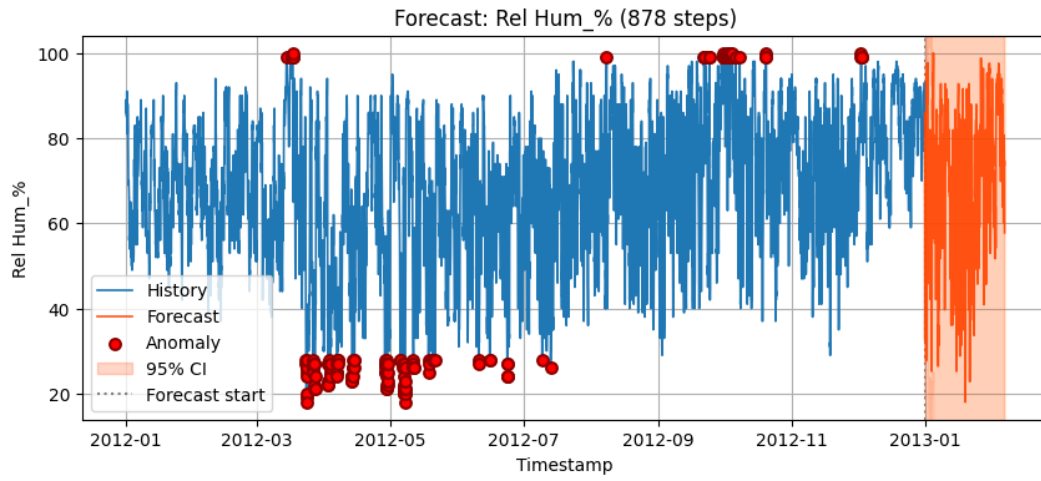
Column: Rel Hum_%

Column: Rel Hum_%



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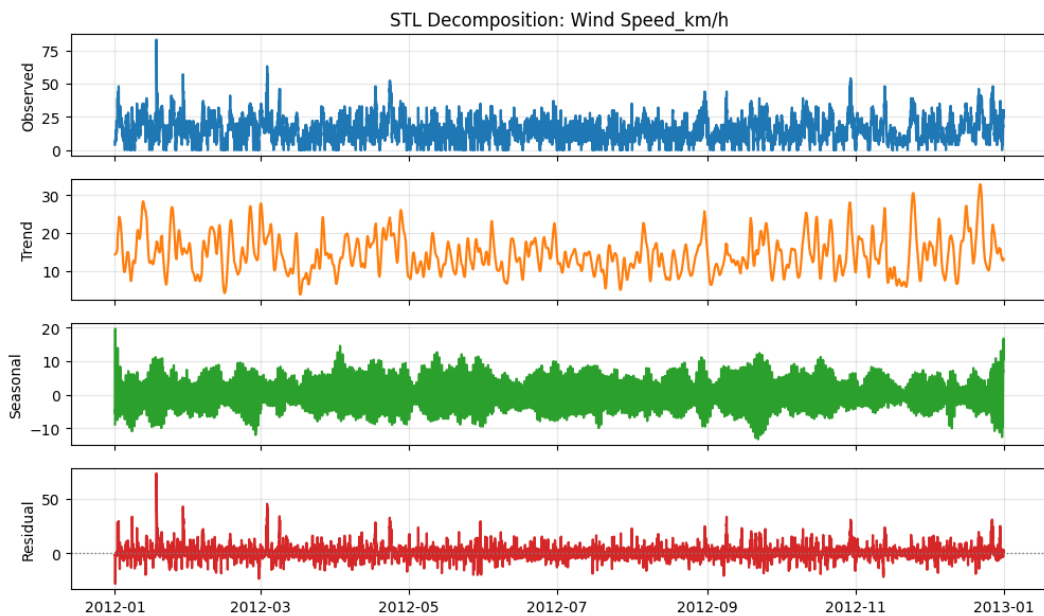
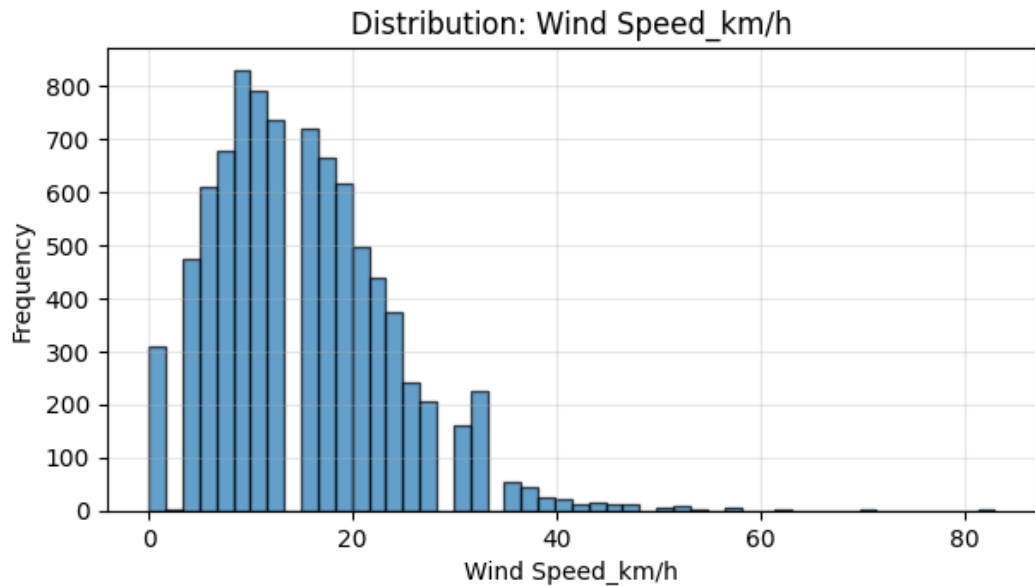


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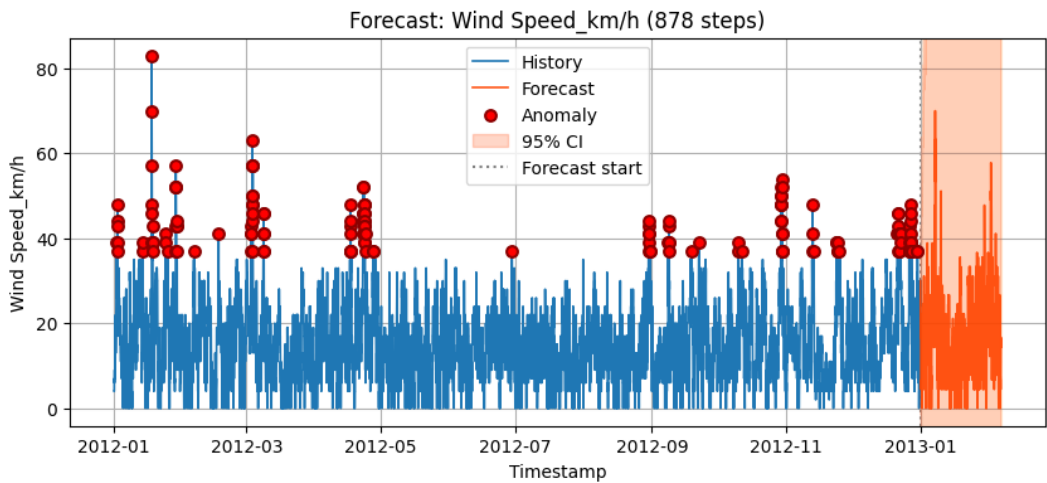
Column: Wind Speed_km/h

Column: Wind Speed_km/h



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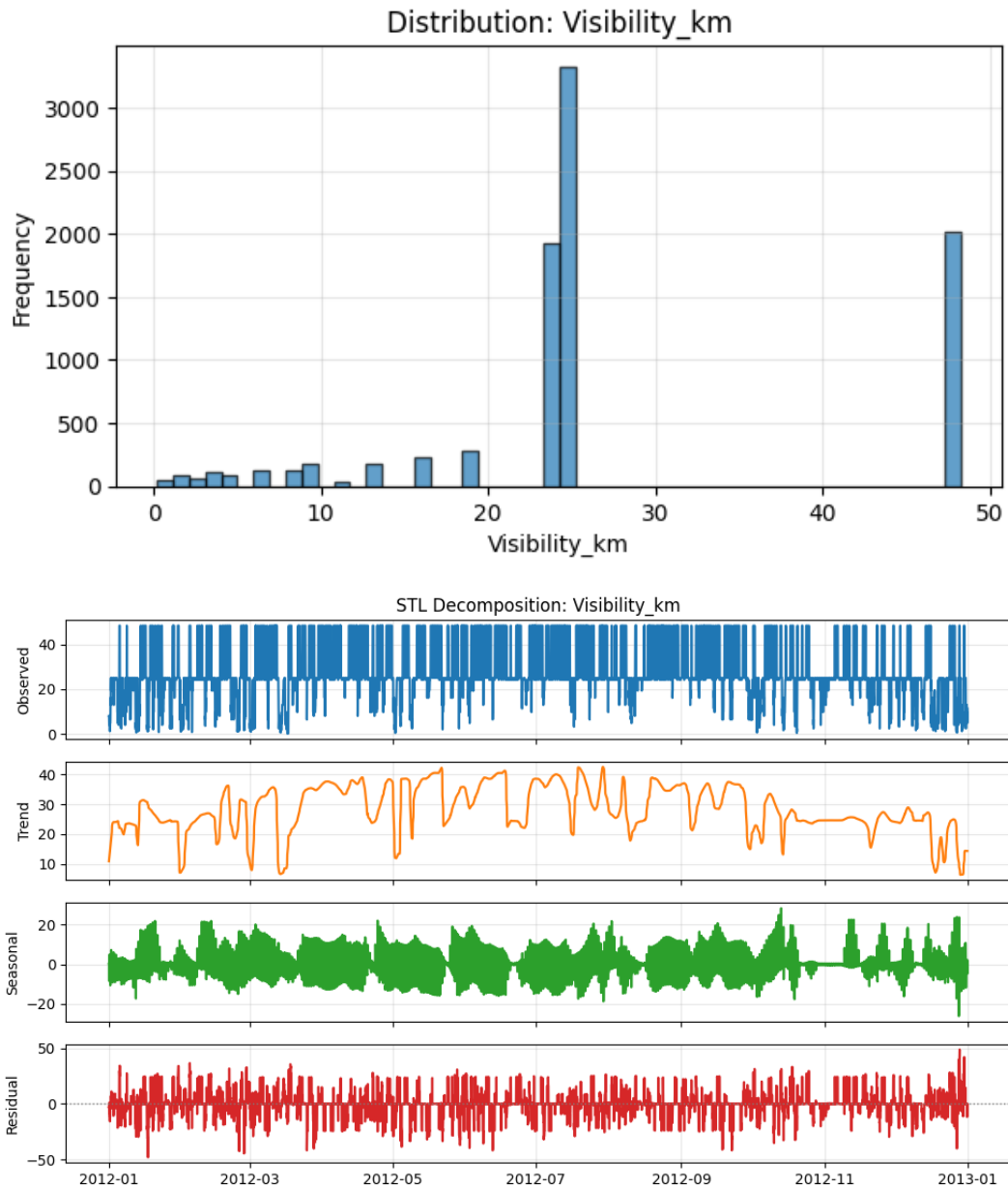


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Dataset: Project_1_-_Weather_Dataset.csv

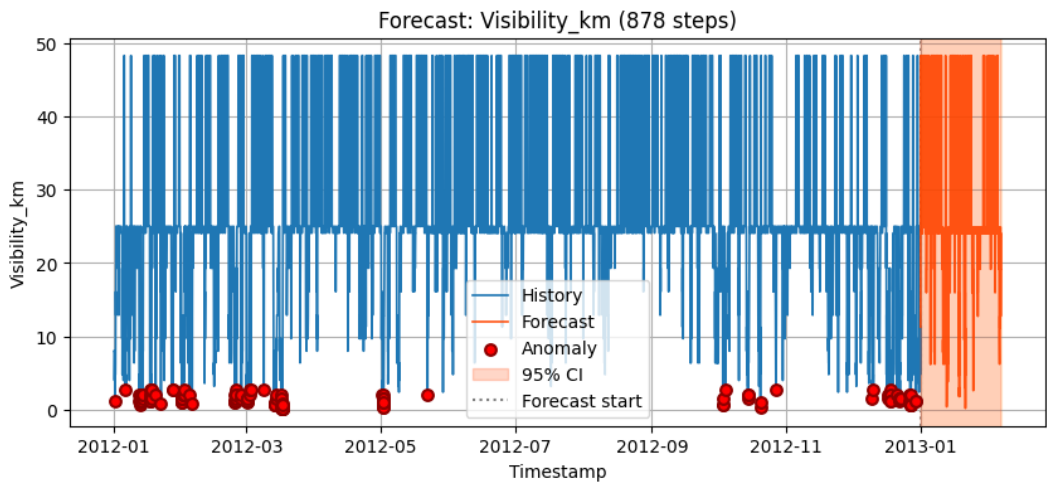
Column: Visibility_km

Column: Visibility_km



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Dataset: Project_1_-_Weather_Dataset.csv

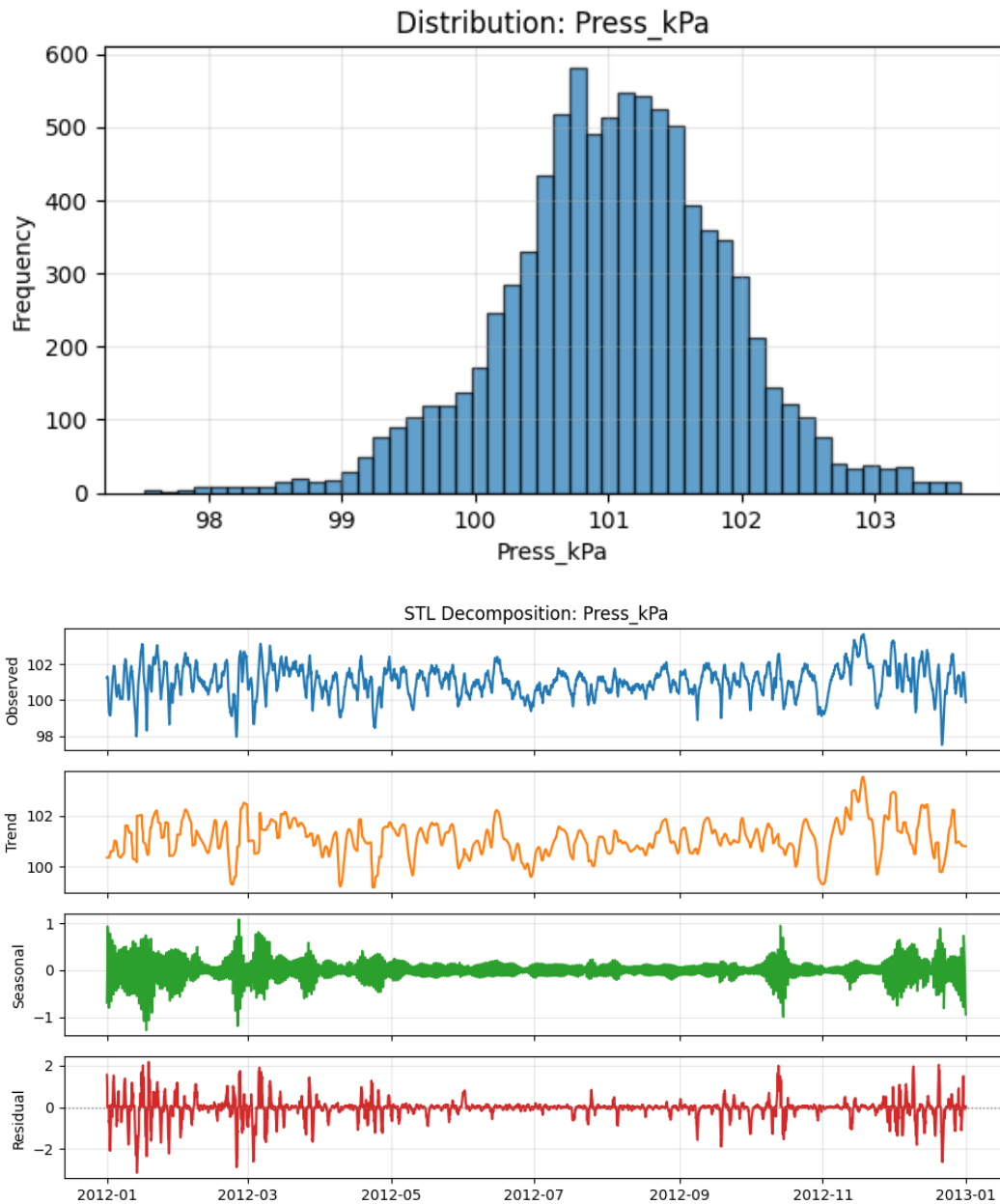


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Dataset: Project_1_-_Weather_Dataset.csv

Column: Press_kPa

Column: Press_kPa



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Dataset: Project_1_-_Weather_Dataset.csv

